

The Feynman Lectures on Physics, Volume II

Problem Set 6

Due 2022.5.29

You are encouraged to discuss the problem sets amongst your peers but the final product you turn in must be done on your own and presented in your own words and calculations. The bonus problems are optional, but I encourage you to do them.

1 Bonus problem: Field Energy and Field Momentum (6pt)

A magnetic field $\mathbf{B}(x)$ points in the $\hat{z}$ direction in half of space as follows.

$$\mathbf{B}(x) = 0, \quad x < 0 \quad (1)$$

$$\mathbf{B}(x) = B_0 \hat{z}, \quad x > 0 \quad (2)$$

There is no electrostatic field. A non-relativistic electron of mass m_e initially moves at constant velocity $\mathbf{v} = v_0 \hat{x}$ at $y = -a$ in the xy plane in the space with $x < 0$. As shown in the figure 1, at $t = 0$ the electron enters the space $x > 0$, travels in a semi-circle of radius a with an angular frequency of $\omega_0 = eB_0/m_e$ for a time π/ω_0 and then exits the region of field with $\mathbf{v} = -v_0 \hat{x}$ at $y = a$.

- (a) What is the differential power per unit solid angle, $dP(t)/d\Omega$, that an observer on the x -axis at a distance $x = r\hat{x} \gg c/\omega_0$ measures when observing the charge? Sketch $dP(t)/d\Omega$ as a function of time and quantitatively label the duration, amplitude, and key features. [*Hint: A lot of progress can be made using a simple model of how a charge radiates.*]
- (b) For an observer on the z -axis at a distance $z = r\hat{z} \gg c/\omega_0$, again sketch the $dP(t)/d\Omega$ he or she would measure as a function of time. Quantitatively label the duration, amplitude, and key features.
- (c) What is the total energy emitted by the electron?

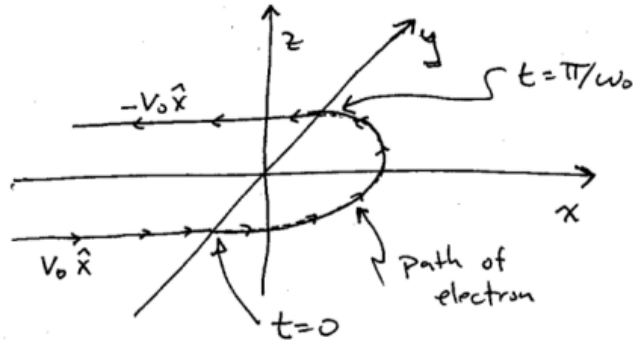


Figure 1:

2 The motion of Charges in Electric and Magnetic Fields (4pt)

A charged particle (charge q , mass m) is initially at rest at the origin. It is acted upon by a constant electric field in the x -direction.

- Calculate the velocity $\mathbf{v}$ and position (x, y, z) of the particle as a function of time (relativistically).
- If the particle has an initial velocity v_0 in the y -direction, how is your answer modified?

3 Refractive Index of Dense Materials (4pt)

In a metal there are plane-wave solutions to Maxwell's equations with the form

$$E_x = E_0 e^{i(\omega t - kz)} \quad (3)$$

where k is a complex number. For low frequencies

$$k = (1 - i) \sqrt{\frac{\sigma \omega}{2\epsilon_0 c^2}} \quad (4)$$

where σ is the conductivity of the metal.

- Write an expression for the magnetic field $\mathbf{B}$ associated with such a wave.
- What is the angle θ between $\mathbf{E}$ and $\mathbf{B}$?
- What is the ratio of the peak value of $\mathbf{B}$ to the peak value of $\mathbf{E}$ at any given value of z ?
- What is the phase difference ϕ between $\mathbf{E}$ and $\mathbf{B}$?

Hint: If the maximum of $\mathbf{E}$ occurs at t_1 and the maximum of $\mathbf{B}$ occurs at t_2 , the phase difference is defined as $\pm\omega(t_1 - t_2)$.

4 Reflection from Surfaces (4pt)

A beam of light with a wavelength $4500\text{\AA}$ (in vacuum) is incident on a prism and totally reflected through 90° , as shown in Fig. 2. The index of refraction of the prism is 1.6.

- (a) Compute the distance d beyond the long side of the prism at which the electric field strength is reduced to $1/e$ of its value just at the surface, assuming the light is polarized so that $\mathbf{E}$ is perpendicular to the plane of incidence.
- (b) Is your answer changed if $\mathbf{E}$ lies in the plane of incidence?

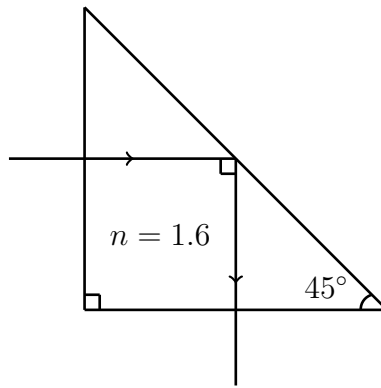


Figure 2: